9.4.1 Handling sessions with Spring Session Data Redis

Spring provides session management features with the Spring Session project. By default, session data is stored in memory, but that's not feasible in a cloud native application. You want to keep it in an external service so that the data survives the application shutdown. Another fundamental reason for using a distributed session store is that you usually have multiple instances of a given application. You'll want them to access the same session data to provide a seamless experience to the user.

Redis is a popular option for session management, and it's supported by Spring Session Data Redis. Furthermore, you have already set it up for the rate limiters. You can add it to Edge Service with minimal configuration.

First you need to add a new dependency on Spring Session Data Redis to the build.gradle file for the Edge Service project. You can also add the Testcontainers library so you can use a lightweight Redis container when writing integration tests. Remember to refresh and reimport the Gradle dependencies after the new addition.

Listing 9.16 Adding dependency for Spring Session and Testcontainers

```
ext {
  ...
  set('testcontainersVersion', "1.17.3")
}

dependencies {
  ...
  implementation 'org.springframework.session:spring-session-data-redis'
  testImplementation 'org.testcontainers:junit-jupiter'
}

dependencyManagement {
  imports {
    ...
    mavenBom
```